



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
HU201 - ENVIRONMENTAL SCIENCE

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. The pollutant primarily responsible for photochemical smog is- (1) CO₂ BL1 PO1
(a) Water vapour (b) Sulphur dioxide
(c) Oxides of nitrogen (d) Ozone
2. BOD test in the laboratory is done for- (1) CO₁ BL1 PO1
(a) 5 days (b) 2 days
(c) 6 days (d) 10 days
3. The cause of landslide is due to- (1) CO₂ BL1 PO1
(a) Deforestation (b) Road construction
(c) Industrialisation (d) All of the above
4. Which one is the intensity level of noise? (1) CO₂ BL1 PO1
(a) Centimetre (b) Decibel
(c) Newton (d) Watt
5. Blackfoot disease is related to- (1) CO₁ BL1 PO1
(a) Hg (b) As
(c) Pb (d) Cd
6. The full form of PAN is (1) CO₁ BL1 PO1
(a) Peroxy acetyl nitrate (b) Peroxy acetyl nitrite
(c) Peroxy acyl nitrate (d) Peroxy acyl nitrite

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| 7. Itai-Itai disease occurs due to | (1) | C02 | BL1 | PO1 |
| (a) Arsenic | (b) Mercury | | | |
| (c) Cadmium | (d) Lead | | | |
| 8. Montreal Protocol is related to | (1) | C03 | BL1 | PO1 |
| (a) Green House effect | (b) Ozone layer Depletion | | | |
| (c) Global warming | (d) Waste water treatment | | | |
| 9. Identify which statement is correct about BOD & COD | (1) | C04 | BL1 | PO2 |
| (a) $BOD < COD$ | (b) $BOD = COD$ | | | |
| (c) $BOD > COD$ | (d) None of these | | | |
| 10. The carrying capacity of an environment refers to- | (1) | C01 | BL1 | PO3 |
| (a) Maximum population size an environment can support | (b) Maximum number of species on Earth | | | |
| (c) Minimum number of individuals needed to survive | (d) Number of births per year | | | |
| 11. Kyoto Protocol is related to- | (1) | C03 | BL1 | PO7 |
| (a) Greenhouse gases | (b) SPM | | | |
| (c) Land pollution | (d) Water pollution | | | |
| 12. Eutrophication in lake is due to the presence of | (1) | C04 | BL1 | PO3 |
| (a) nitrogen and phosphorous | (b) DO and Bacteria | | | |
| (c) Chromium and Mercury | (d) Cd | | | |

Part B**(Answer any four of the following) 5x4=20**

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| 13. What is Global Warming? Analyze the causes and comment on the proposed remedial steps. | (5) | C03 | BL4 | PO7 |
| 14. (a) What is acid rain? | (1) | C04 | BL1 | PO2 |
| (b) Write the chemical reactions involved in it and state effects of acid rain. | (4) | C04 | BL3 | PO2 |
| 15. Discuss about heavy metal toxicity for Arsenic (As) and Mercury (Hg). | (5) | C04 | BL4 | PO1 |
| 16. Prove that maximum sustainable yield obtained when population is half of carrying capacity ($N = K/2$) | (5) | C02 | BL4 | PO1 |

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| 17. Write short note on Electrostatic precipitator (ESP) | (5) | C02 | BL2 | PO1 |
| 18. Differentiate between Photochemical smog and London smog. | (5) | C02 | BL1 | PO1 |

Part C
(Answer any three of the following) 3x10=30

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| 19. (a) Establish the relation $BOD_t = L_0 (1 - e^{-kt})$ | (5) | C04 | BL4 | PO3 |
| (b) Distinguish between Pollutant and Contaminant. | (3) | C01 | BL2 | PO1 |
| (c) What is the difference between climate and weather? | (2) | C01 | BL2 | PO3 |
| 20. (a) Derive the equation for exponential growth pattern. | (4) | C01 | BL3 | PO3 |
| (b) Write a short note on environmental degradation. | (3) | C01 | BL2 | PO3 |
| (c) The increase in population from 1 million to 10 million took 200 years for exponential growth at constant rate. Find out the growth rate. | (3) | C01 | BL4 | PO3 |
| 21. (a) Write down all green house gases. Define Primary and Secondary air pollutants with examples | (5) | C03 | BL2 | PO7 |
| (b) Write a short note on Environmental Impact Assessment (EIA). | (5) | C04 | BL3 | PO7 |
| 22. (a) What is hardness of water? How temporary hardness and permanent hardness of water can be removed ? | (6) | C03 | BL3 | PO3 |
| (b) A waste sample contains 50 mg/l of Ca^{2+} , 36.5mg/l of Mg^{2+} . What is the total hardness in mg/l as $CaCO_3$? | (4) | C04 | BL4 | PO3 |
| 23. (a) Discuss the biochemical effect of Mercury with possible reaction mechanism and translocation of mercury in food chain. | (5) | C02 | BL2 | PO1 |
| (b) What is doubling time for population? Derive the mathematical expression. | (5) | C01 | BL3 | PO1 |
- The human population follows exponential growth rate of 2% per year. In the year 1980, the world population was 5 billion. When will the population reach at 12 billion?